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Characterization of Precipitate Reactor Feed Tank (PRFT) Batches 15, 22, and 26 from the Defense Waste Processing Facility (DWPF)

S. C. Hunter

November 2024

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EXECUTIVE SUMMARY

The Savannah River Site (SRS) Defense Waste Processing Facility (DWPF) processes a Monosodium Titanate/Sludge Solids (MST/SS) waste stream received from the Salt Waste Processing Facility (SWPF) via the Precipitate Reactor Feed Tank (PRFT). During processing, DWPF is required to provide evidence of compliance with the Waste Acceptance Product Specifications (WAPS) to ensure acceptance of their vitrified high-level waste (HLW) into the Civilian Radioactive Waste Management System. Production Records must document the constituents of the MST/SS material in the PRFT from each salt batch (StB) processed at SWPF. Savannah River Mission Completion (SRMC) has requested Savannah River National Laboratory (SRNL) to analyze PRFT samples representing each SWPF salt batch for thirty-two radionuclides.

Additionally, elemental analysis of PRFT slurry and MST/SS solids was performed to aid SRMC in further refinement of the inputs and assumptions used in future frit development and Material Tracking Program calculations.

The analyses of PRFT Batches 15, 22, and 26, which corresponds to material from the processing of StB4, StB5, and StB7, respectively, are reported herein. The unwashed dried solids of the PRFT batches were found to be 86-87% MST. Additionally, the total sulfur values are well below the assumed 982 mg of sulfate/kg of PRFT slurry used in Material Tracking Program calculations.

TABLE OF CONTENTS

LIST OF TABLES viii
LIST OF FIGURES viii
LIST OF ABBREVIATIONS ix
1.0 Introduction 1
2.0 Experimental Procedure 2
2.1 Transferring and Filtering PRFT Samples 2
2.2 Wt % Solids and Density 3
2.3 Chemical Analyses 3
2.3.1 PRFT Dried Unwashed Solids 3
2.3.2 PRFT Slurries 4
2.4 Quality Assurance 4
3.0 Results and Discussion 4
3.1 Density and Weight Percent Solids 4
3.2 PRFT Slurry Elemental Composition 5
3.3 Elemental Composition of the Dried Unwashed Solids 7
3.4 PRFT MST/SS Radionuclide Results 9
3.5 Slurry and MST/SS Solids Oxide Conversion 15
4.0 Conclusions 16
5.0 References 17
Appendix A. Comparison of Activities between Slurry and Dried Unwashed Solids ICP-MS Data 19
Appendix B. Comparison of PRFT Batches to SSRT Composition Estimates 20

LIST OF TABLES

Table 3-1. PRFT Sample Densities and Wt % Solids 5

Table 3-2. Elemental Composition of the PRFT Slurry 6

Table 3-3. Estimate of the Amount of MST in the Total Dried Solids 7

Table 3-4. Elemental Analysis of Dried Unwashed Solids 8

Table 3-5. Estimate of the Percentage of MST in the Dried Unwashed Solids 9

Table 3-6. Elemental Composition of the Dried Unwashed Solids with MST Excluded. Elements given are >0.1 wt % in at least one PRFT Batch 9

Table 3-7. Radionuclide Results for PRFT Batch 15 Dried Unwashed Solids 11

Table 3-8. Radionuclide Results for PRFT Batch 22 Dried Unwashed Solids 12

Table 3-9. Radionuclide Results for PRFT Batch 26 Dried Unwashed Solids 13

Table 3-10. Tc-99 Results Comparison 14

Table 3-11. U-235 Enrichment of the PRFT Slurry and StB Unfiltered Salt Solution 14

Table 3-12. Uranium and Plutonium Fissile Isotopes 15

Table 3-13. Oxide Conversion for Elements in PRFT Batches 16

LIST OF FIGURES

Figure 2-1. Photographs of PRFT Slurry A) Batch 15, B) Batch 22, and C) Batch 26 3

LIST OF ABBREVIATIONS

AR	Aqua Regia
ATFC	Analytical and Tank Farm Characterization
DMA	Direct Mercury Analyzer
DWPF	Defense Waste Processing Facility
HLW	High-Level Waste
IAEA	International Atomic Energy Agency
ICP-ES	Inductively Coupled Plasma-Emission Spectroscopy
ICP-MS	Inductively Coupled Plasma-Mass Spectrometry
ISS	Insoluble Sludge Solids
MAR	Measurement Acceptance Region
MDA	Minimum Detectable Activity
MST/SS	Monosodium Titanate/Sludge Solids
n	Sample Size
NM	Nuclear Measurements
PF	Peroxide Fusion
PRFT	Precipitate Reactor Feed Tank
RSD	Relative Standard Deviation
SRMC	Savannah River Mission Completion
SRNL	Savannah River National Laboratory
SRS	Savannah River Site
SSRT	Sludge Solids Receipt Tank
StB	Salt Batch
SWPF	Salt Waste Processing Facility
TTQAP	Task Technical and Quality Assurance Plan
TTR	Technical Task Request
WAPS	Waste Acceptance Product Specifications
WQR	Waste Form Qualification Report

1.0 Introduction

The Savannah River Site (SRS) Defense Waste Processing Facility (DWPF) processes a Monosodium Titanate/Sludge Solids (MST/SS) waste stream received from the Precipitate Reactor Feed Tank (PRFT) via the Sludge Solids Receipt Tank (SSRT) at the SRS Salt Waste Processing Facility (SWPF). During processing, DWPF is required to provide evidence of compliance with the Waste Acceptance Product Specifications (WAPS)[1] to ensure acceptance of their vitrified high-level waste (HLW) into the Civilian Radioactive Waste Management System. This evidence is documented in the form of Production Records for DWPF canistered waste forms.

Production Records must document the constituents of the MST/SS material in the PRFT from each salt batch (StB) processed at SWPF. In accordance with the Waste Form Qualification Report (WQR)[2] for glass qualification, all radionuclides with a half-life longer than ten years and a curie concentration within the glass product of at least 0.01% of the total radionuclide activity up to 1,100 years after production must be reported. Within the WQR, a minimum of 30 radionuclides are reportable; however, this list can be expanded or reduced based on the projected curie content. In conjunction with International Atomic Energy Agency (IAEA) requirements, there are also other uranium and plutonium isotopes that must be included in the HLW analyses (e.g., U-235 and U-236). Since titanium is utilized within the MST/SS stream, its elemental analysis is used to determine the dilution factor for the MST/SS radionuclide calculation in the glass product. The analysis of the fissile uranium and plutonium isotopes and elements Fe and Mn (neutron poisons) in the PRFT samples also provides useful nuclear criticality safety data for the calculation of poison-to-fissile ratios in the MST/SS stream. Savannah River Mission Completion (SRMC) has requested Savannah River National Laboratory (SRNL) to analyze PRFT samples on each SWPF salt batch for the following thirty-two radionuclides.

Radionuclides	
Nickel-59	Neptunium-237
Nickel-63	Plutonium-238
Selenium-79	Plutonium-239
Strontium-90	Plutonium-240
Zirconium-93	Plutonium-241
Niobium-93m	Plutonium-242
Technetium-99	Americium-241
Tin-126	Americium-242m
Cesium-137	Americium-243
Samarium-151	Curium-244
Thorium-229	Curium-245
Uranium-233	Curium-246
Uranium-234	Curium-247
Uranium-235	Curium-248
Uranium-236	Californium-249
Uranium-238	Californium-251

The analysis of PRFT batches could also aid in further refinement of inputs and assumptions for frit development calculations. Currently, during sludge batch qualification, iterative Measurement Acceptance Region (MAR) assessments are performed based on projected compositions to identify acceptable frits for processing. The composition of insoluble sludge solids (ISS) in the MST/SS stream has not been previously measured during salt batch qualification. Instead, MAR assessments assumed the entrained ISS to be the composition of Sludge Batch 9 (adjusted to remove soluble species).[3]

In Material Tracking Program calculations, the sulfate solubility limit in glass is protected by determining a conservative permissible waste loading based on sample concentrations in sludge and assumed concentrations in the MST/SS stream. While soluble sulfate is measured during salt batch qualification, insoluble sulfur compounds currently are not. Analysis of the as-received PRFT samples, particularly total sulfur, will aid SRMC in refining the assumed sulfate concentration of the MST/SS stream.

SRNL initially received PRFT Batches 15, 22, and 26, which correspond to material from the processing of StB4, StB5, and StB7, respectively. The analysis of PRFT batches corresponding to earlier salt batches will be documented in a future report. Reported herein are the analyses of the initial received PRFT Batches 15, 22, and 26.

2.0 Experimental Procedure

2.1 Transferring and Filtering PRFT Samples

SRMC pulled PRFT batches 15, 22, and 26 on 04/27/22, 10/29/22, and 02/17/23, respectively. The PRFT samples were received at SRNL on March 2, 2023. The PRFT slurries were transferred from the 200 mL stainless steel dip bottles into 250 mL high density polyethylene bottles. The dip bottles were washed with the supernate of the PRFT samples until there were no visible solids in the dip bottles. The MST/SS solids were collected by filtering a portion of the PRFT slurries through a 0.45 um nylon filter. The MST/SS solids were not washed prior to being dried at 110°C overnight. Photos of the three PRFT slurries, Figure 2-1, were taken after transferring the slurries out of the dip bottles. As can be seen from the images, PRFT Batch 26 has a peachy color, compared to the white milky color of the other PRFT batches. The different color is most likely due to the use of a sludge simulant filter aid during the processing of StB7 at SWPF.[4]

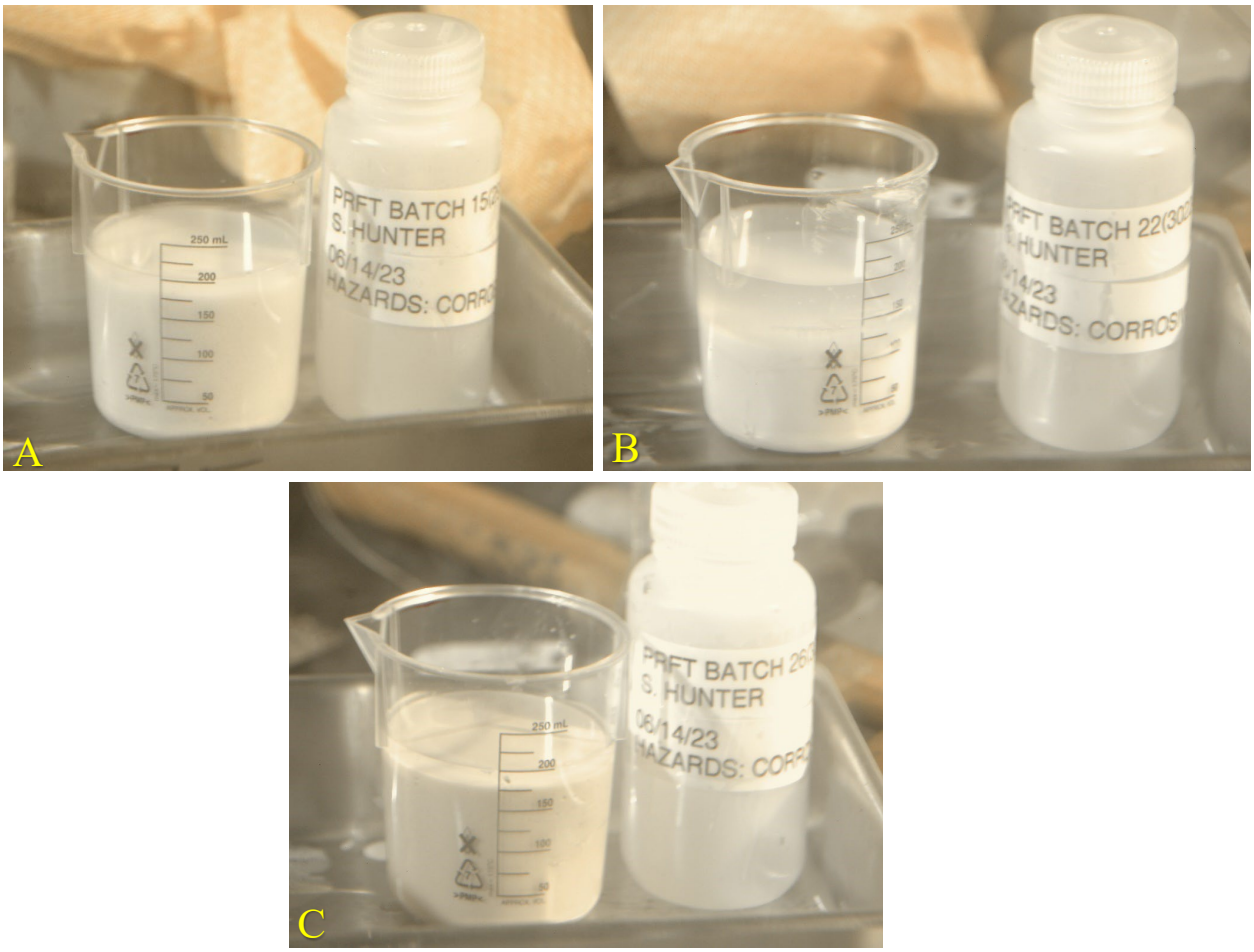


Figure 2-1. Photographs of PRFT Slurry A) Batch 15, B) Batch 22, and C) Batch 26

2.2 Wt % Solids and Density

The slurry densities were determined gravimetrically from sample weights in vessels of known volume. Representative PRFT supernatant samples were obtained by filtering a portion of the slurries through a 0.45 um nylon filter. The weight percent total solids and weight percent dissolved solids were determined by drying aliquots of the slurry and filtrate at 110 °C until a constant weight was observed. Weight percent insoluble and soluble solids were calculated using Equation 1 and 2.

$$Wt\ \%_{insoluble} = \frac{(Wt\ \%_{total} - Wt\ \%_{dissolved})}{(100 - Wt\ \%_{dissolved})} \times 100\% \quad (1)$$

$$Wt\ \%_{soluble} = Wt\ \%_{total} - Wt\ \%_{insoluble} \quad (2)$$

2.3 Chemical Analyses

2.3.1 PRFT Dried Unwashed Solids

The PRFT dried unwashed solids were digested, in quadruplicate, by HNO_3/HCl (aqua regia, AR)[5] with added HF in sealed Teflon vessels. The digested sample was diluted to 100 mL with deionized water. Digested samples were pulled from these dilutions and submitted to Analytical and Tank Farm Characterization (ATFC) for Inductively Coupled Plasma-Emission Spectroscopy (ICP-ES) and

Inductively Coupled Plasma-Mass Spectrometry (ICP-MS). They were also submitted to Nuclear Measurements (NM) for the following methods: gamma counting (Cs-137), Cs-removed gamma counting (Sn-126), Se-79 method (Se-79), Sr-90 method (Sr-90), Ni-59/-63 method (Ni-59 and Ni-63), Pm-147/Sm-151 method (Sm-151), Th-229/-230 method (Th-229), Tc-99 method (Tc-99), and Pu-238/-241 Method (Pu-238 and Pu-241). To get better detection limits, dried unwashed MST/SS solids were directly submitted to NM for Am/Cm method (Am-241, Am-242m, Am-243, Cm-244/-245/-246/-247/-248, Cf-249, and Cf-251). These methods have been described previously.[6]

The dried unwashed solids were also digested, in quadruplicate, by $NaOH/Na_2O_2$ (peroxide fusion, PF) using Zr crucibles.[7] PF digested samples of the dried unwashed solids were diluted with deionized water and samples were submitted to ATFC for ICP-ES analysis.

2.3.2 PRFT Slurries

The PRFT slurries were digested, in quadruplicate, by AR/HF. The digested sample was diluted to 100 mL with deionized water. Digested samples were pulled from these dilutions and submitted to ATFC for ICP-ES, ICP-MS, and Direct Mercury Analysis (DMA). Slurry samples were also digested, in quadruplicate, by PF.[7] The slurry samples were diluted to 100 mL with deionized water and submitted to ATFC for ICP-ES analysis.

Co, Nd, Pb, Sb, and U were determined by ICP-MS. Co was calculated from mass 59. Nd was calculated from the sum of masses 143 through 146, 148, and 150. Mass 142 was not included due to the potential interference from Ce-142. Pb was determined from the sum of masses 206, 207, and 208. Sb was determined from the sum of masses 121 and 123. U was calculated from the sum of masses 233-236 and 238.[8]

The one sigma analytical uncertainty reported by ATFC was 10% for ICP-ES, DMA, and ICP-MS.

2.4 Quality Assurance

This work was performed under a Technical Task Request (TTR).[9] The analysis herein satisfies all three tasks of the Task Technical and Quality Assurance Plan (TTQAP) associated with this TTR.[10] The TTR identifies the radionuclide results (Task 1 of the TTQAP) as waste form affecting and complies with the requirements of RW-0333P. The TTR identifies the Functional Classification as Safety Class. Thus, this document was reviewed by Design Verification by Document Review. The requirements for performing reviews of technical reports and the extent of review are established in manual E7 2.60. SRNL documents the extent and type of review using the SRNL Technical Report Design Checklist contained in WSRC-IM-2002-00011, Rev. 2. Data are recorded in the electronic laboratory notebook K7482-00430-29.

3.0 Results and Discussion

3.1 Density and Weight Percent Solids

The slurry densities, as well as the weight percent solids measurements, for the PRFT batches are given in Table 3-1.

Table 3-1. PRFT Sample Densities and Wt % Solids

Property	PRFT Batch 15		PRFT Batch 22		PRFT Batch 26	
	Result	% RSD, n	Result	% RSD, n	Result	% RSD, n
Wt % Total Solids (Slurry Basis)	4.14	2.0, 4	5.04	3.9, 4	4.86	2.4, 4
Wt % Dissolved Solids (Supernatant Basis)	1.34	7.7, 4	1.17	8.3, 4	0.94	5.3, 4
Wt % Insoluble Solids (Slurry Basis)	2.84	NA	3.92	NA	3.96	NA
Wt % Soluble Solids (Supernatant Basis)	1.3	NA	1.12	NA	0.90	NA
Slurry Density (g/mL)	1.031	0.32, 4	1.035	0.41, 4	1.033	0.18, 4

RSD = relative standard deviation
n = sample size

3.2 PRFT Slurry Elemental Composition

The elemental composition results for the PRFT slurries are given in Table 3-2. The elements were determined from ICP-ES and ICP-MS. Results are given in mg/kg slurry. The data for Al and Si from the PF digestion is reported since some Al and Si species may not be soluble in an AR digestion. Due to the use of Zr crucibles and sodium-containing reagents ($NaOH/Na_2O_2$) in PF digestions, both Zr and Na values were determined from the AR/HF digestion. For all other elements quantified by ICP-ES, all digestion results were averaged if the difference in the results from the two digestion methods was 10% or less. If not within 10%, the results are taken from the digestion with the larger values. When all results are below the detection limit, the minimum replicate result is used. The Ca values may be biased high due to Ca in the blank. For PRFT Batch 15, two of the AR replicates did not have added HF. For those two replicates, only Ti and S had lower values compared to the replicates with added HF.

Titanium and sodium are the predominate elements in the slurry. The reported sulfur, in mg/kg of slurry, for PRFT Batch 15, 22, and 26 is 99.4, 80.4, and 66.9, respectively. If you assume that all the sulfur is from sulfate you would get 298, 241, and 200 mg/kg for PRFT batch 15, 22, and 26. These values are well below the assumed 982 mg/kg of sulfate in the MST/SS stream.[11] The values are in line with the DWPF lab's sulfate values for PRFT batch 15, 22, and 26, which are <579, <580, and <606 ppm.[12] The total mercury of PRFT Batch 26 was also determined from the acid-digested slurry using DMA. Run in duplicate, the mercury value obtained was 403 mg/kg of slurry (% RSD = 0.7).

Table 3-2. Elemental Composition of the PRFT Slurry

Element	PRFT Batch 15		PRFT Batch 22		PRFT Batch 26	
	mg/kg slurry	% RSD, n	mg/kg slurry	% RSD, n	mg/kg slurry	% RSD, n
Ag	*<0.077*	*NA*	*<0.077*	*NA*	*<0.077*	*NA*
Al	**219**	3.8, 4	**162**	4.2, 4	**283**	7.1, 4
B	*<8.78*	*NA*	*<8.78*	*NA*	*<8.78*	*NA*
Ba	**10.2**	4.6, 8	**12.4**	1.0, 8	**14.5**	1.4, 8
Be	*<0.0505*	*NA*	*<0.0503*	*NA*	*<0.0504*	*NA*
Ca	**195**	2.7, 4	**197**	5.3, 4	**247**	5.1, 4
Cd	**7.44**	4.6, 8	**7.55**	1.4, 4	**6.61**	0.29, 4
Ce	*<0.305*	*NA*	*<0.305*	*NA*	*<0.305*	*NA*
Co†	**7.13**	5.6, 4	**2.35**	2.1, 4	**1.81**	2.1, 4
Cr	**16.1**	4.3, 4	**17.9**	15, 4	**16.5**	26, 4
Cu	*<17.8*	*NA*	*<17.8*	*NA*	*<17.8*	*NA*
Fe	**154**	3.2, 4	**140**	8.6, 8	**479**	6.6, 8
Gd	*<0.342*	*NA*	*<0.342*	*NA*	*<0.342*	*NA*
K	*<26.1*	*NA*	*<26.1*	*NA*	*<26.1*	*NA*
La	*<0.291*	*NA*	*<0.291*	*NA*	*<0.291*	*NA*
Li	*<61.6*	*NA*	*<61.6*	*NA*	*<61.6*	*NA*
Mg	**7.82**	7.1, 4	**7.19**	26, 4	**6.37**	2.9, 3
Mn	**5.40**	4.6, 8	**2.24**	3.8, 8	**42.2**	1.4, 8
Mo	*<1.76*	*NA*	*<1.76*	*NA*	*<1.76*	*NA*
Na	**6558**	2.3, 4	**6723**	0.87, 4	**6138**	0.92, 4
Nd†	*< 0.199*	*NA*	*< 0.199*	*NA*	*< 0.199*	*NA*
Ni	**19.4**	4.3, 8	**20.2**	5.5, 8	**50.4**	1.2, 8
P	*<6.81*	*NA*	*<6.79*	*NA*	*<6.81*	*NA*
Pb†	**6.11**	2.5, 4	**4.01**	0.43, 4	**6.17**	0.19, 4
S	**99.4**	0.43, 2	**80.4**	2.5, 4	**66.9**	2.2, 4
Sb†	*< 0.0398*	*NA*	*< 0.0398*	*NA*	*< 0.0398*	*NA*
Si	*<43.9*	*NA*	*<43.9*	*NA*	*<43.9*	*NA*
Sn	*<4.89*	*NA*	*<4.88*	*NA*	*<4.89*	*NA*
Sr	**4.69**	3.1, 8	**3.10**	4.7, 8	**4.46**	3.7, 8
Th	*<5.52*	*NA*	*<5.52*	*NA*	*<5.52*	*NA*
Ti	**12200**	3.7, 6	**16200**	2.4, 8	**16225**	1.6, 8
U*	**133**	4.9, 4	**141**	1.8, 4	**129**	0.4, 4
V	*<2.98*	*NA*	*<2.97*	*NA*	*<2.98*	*NA*
Zn	**12.4**	6.2, 8	**15.7**	7.7, 7	**23.4**	26, 4
Zr	**4.99**	9.8, 4	**6.92**	0.83, 4	**6.90**	0.99, 4

† Calculated from MS data for Co: mass 59; Nd: sum of masses 143-146, 148 and 150; Pb: sum of masses 206-208; Sb: sum of masses 121 and 123.
* Calculated from MS data for sum of masses 233-236 and 238. U from ICP-ES was very similar for the three batches.

The relative percentage of MST can be estimated using the titanium values given in Table 3-2 and with the ratio of 0.479 g of Ti/g of MST. Table 3-3 provides these estimates, assuming that all Ti present in the PRFT slurries is from the MST.

Table 3-3. Estimate of the Amount of MST in the Total Dried Solids

	PRFT Batch 15	PRFT Batch 22	PRFT Batch 26
MST (%)	62	67	70

3.3 Elemental Composition of the Dried Unwashed Solids

The elemental analysis of the MST/SS that were collected on a filter and dried is given in Table 3-4. The majority of the elements in the table are from the ICP-ES data from one or both of the digestions. ICP-MS was used for Co, Pb, Sb, and U as noted in the footnote. As expected, the solids are predominately made up of sodium and titanium from the MST in the solids. There is also a significant amount of aluminum, calcium, and iron. As described above for the PRFT slurries, the amount of iron and titanium can be used to estimate the amount of sludge and MST. Table 3-5 gives the estimate for MST in the dried unwashed solids. The MST estimate can be subtracted from the total unwashed dried solids to give the significant (>0.5 wt %) elemental contributors of the entrained sludge solids. These are given in Table 3-6. It should be noted that all the uranium given in the table may be adsorbed to the MST. The higher amount of Fe and Mn in PRFT batch 26 appears to correlate with the use of a sludge simulant filter aid at SWPF during StB7 processing.

Table 3-4. Elemental Analysis of Dried Unwashed Solids

Element	PRFT Batch 15		PRFT Batch 22		PRFT Batch 26	
	Wt %	% RSD, n	Wt %	% RSD, n	Wt %	% RSD, n
Ag	*< 1.60E-04*	*NA*	*< 1.71E-04*	*NA*	*< 1.64E-04*	*NA*
Al	**3.98E-01**	11, 4	**2.45E-01**	17, 4	**5.95E-01**	16, 4
B	*< 1.21E-02*	*NA*	*< 1.21E-02*	*NA*	*< 1.21E-02*	*NA*
Ba	**3.48E-02**	3.7, 8	**3.14E-02**	4.2, 8	**3.76E-02**	10, 8
Be	*< 1.05E-04*	*NA*	*< 1.12E-04*	*NA*	*< 1.07E-04*	*NA*
Ca	**2.95E-01**	4.8, 4	**2.46E-01**	5.3, 4	**3.19E-01**	3.4, 4
Cd	**2.65E-02**	4.0, 8	**1.88E-02**	5.3, 8	**1.66E-02**	9.7, 8
Ce	*< 4.05E-03*	*NA*	*< 4.33E-03*	*NA*	*< 4.14E-03*	*NA*
Co†	**2.77E-02**	1.6, 4	**5.72E-03**	3.0, 4	**4.26E-03**	2.1, 4
Cr	**3.54E-02**	2.3, 4	**2.69E-02**	5.2, 4	**2.10E-02**	1.6, 4
Cu	**2.52E-02**	11, 4	**3.04E-02**	3.5, 4	**3.10E-02**	10, 4
Fe	**4.93E-01**	3.8, 8	**3.60E-01**	5.2, 8	**1.26E+00**	9.9, 8
Gd	*< 1.61E-03*	*NA*	*< 1.67E-03*	*NA*	*< 1.55E-03*	*NA*
K	*< 5.13E-02*	*NA*	*< 5.48E-02*	*NA*	*< 5.24E-02*	*NA*
La	*< 5.18E-04*	*NA*	*< 5.54E-04*	*NA*	*< 5.30E-04*	*NA*
Li	*< 2.04E-01*	*NA*	*< 2.18E-01*	*NA*	*< 2.09E-01*	*NA*
Mg	**2.36E-02**	2.7, 4	**1.22E-02**	7.9, 4	**1.53E-02**	13, 4
Mn	**1.84E-02**	5.3, 8	**6.62E-03**	3.0, 4	**1.07E-01**	10, 8
Mo	*< 3.66E-03*	*NA*	*< 3.91E-03*	*NA*	*< 3.74E-03*	*NA*
Na	**1.08E+01**	1.4, 4	**1.03E+01**	5.5, 4	**9.82E+00**	9.2, 4
Ni	**6.65E-02**	5.3, 8	**5.19E-02**	6.0, 8	**1.32E-01**	8.9, 8
P	*< 4.11E-02*	*NA*	*< 4.39E-02*	*NA*	*< 4.20E-02*	*NA*
Pb†	**2.07E-02**	2.0, 4	**1.09E-02**	18, 4	**1.52E-02**	1.2, 4
S	*< 5.37E-02*	*NA*	*< 5.41E-02*	*NA*	*< 5.41E-02*	*NA*
Sb†	*< 1.99E-04*	*NA*	*< 2.13E-04*	*NA*	*< 2.03E-04*	*NA*
Si	*< 1.02E-01*	*NA*	*< 1.02E-01*	*NA*	*< 1.02E-01*	*NA*
Sn	*< 1.47E-02*	*NA*	*< 1.57E-02*	*NA*	*< 1.50E-02*	*NA*
Sr	**1.71E-02**	3.2, 8	**8.85E-03**	5.8, 8	**1.24E-02**	10, 8
Th	*< 5.10E-02*	*NA*	*< 5.46E-02*	*NA*	*< 5.22E-02*	*NA*
Ti	**4.15E+01**	2.7, 8	**4.12E+01**	3.8, 8	**4.17E+01**	8.8, 8
U*	**4.50E-01**	1.9, 4	**3.50E-01**	2.4, 4	**3.28E-01**	3.4, 4
V	*< 6.19E-03*	*NA*	*< 6.61E-03*	*NA*	*< 6.32E-03*	*NA*
Zn	**3.90E-02**	3.6, 8	**3.76E-02**	5.2, 7	**4.90E-02**	7.1, 8
Zr	**1.88E-02**	2.8, 4	**1.81E-02**	3.3, 4	**1.77E-02**	2.4, 4

† Calculated from MS data for Co: mass 59; Pb: sum of masses 206-208; Sb: sum of masses 121 and 123.
* Calculated from MS data for sum of masses 233-236 and 238. U from ICP-ES was very similar for the three batches.

Table 3-5. Estimate of the Percentage of MST in the Dried Unwashed Solids

	PRFT Batch 15	PRFT Batch 22	PRFT Batch 26
MST (%)	87	86	87

Table 3-6. Elemental Composition of the Dried Unwashed Solids with MST Excluded. Elements given are >0.1 wt % in at least one PRFT Batch

	Wt % of Unwashed Dried Solids w/MST Excluded		
	PRFT Batch 15	PRFT Batch 22	PRFT Batch 26
Al	3.0	1.7	4.5
Ba	0.26	0.22	0.29
Ca	2.2	1.7	2.4
Cd	0.20	0.13	0.13
Co	0.21	0.04	0.03
Cr	0.26	0.19	0.16
Cu	0.19	0.21	0.24
Fe	3.7	2.5	9.6
Mg	0.18	0.09	0.12
Mn	0.14	0.05	0.82
Na	6.42	3.29	0.00
Ni	0.49	0.37	1.01
Sr†	0.13	0.06	0.09
U†	3.3	2.5	2.5
Zn	0.29	0.27	0.37
Zr	0.14	0.13	0.14

† Uranium, along with other actinides, and strontium may be adsorbed to the MST solids.

3.4 PRFT MST/SS Radionuclide Results

The radionuclide compositions of the PRFT Batches 15, 22, and 26 are given in Tables 3-7, 3-8, and 3-9. The tables give the activity in µCi/g of unwashed MST/SS dried solids, the activity in pCi/g of Ti, and the associated relative standard deviation of the replicate analyses for each radionuclide. The percent relative standard deviations are given for quadruplicate analyses unless otherwise indicated. Activity measurements that were below the detection limit of a given technique or calculated do not have an associated relative standard deviation and are marked as 'NA' in the tables. The analytical uncertainty and method for each of the radionuclides is also shown. The minimum detectable activity (MDA) is the value above which the instrument signal can be considered quantitative relative to the signal-to-noise ratio. The upper limit is a measurable value above the MDA that is biased high due to spectral interference or blank contamination.

Nb-93m was calculated as previously described using the Zr-93 activity and an estimated waste age of 55 years for all three PRFT batches.[6]

Strontium and most actinides are known to absorb to MST. PRFT Batch 26 has a higher concentration of Sr-90 compared to the other two batches. While this does not take into account heels in the various tanks, the results appear to align with salt batch qualification results where StB7 also has a higher concentration

of Sr-90 compared to StB5 and StB4. There does not appear to be a similar relationship between the PRFT batches and their corresponding salt batch for U and Pu radionuclide results. SWPF has recently changed to MST strikes as low as 0.1 g/L; future PRFT batch results could be compared in more detail to StB qualification and Tank 50 data to determine the efficiency of the MST strikes at SWPF. Additionally, elements such as Am and Zr have the potential to absorb depending on the alkalinity of the solution and solubility.[13,14] Cesium and technetium would not be expected to absorb to MST. The cesium activity is likely from residual supernate that remained as interstitial liquid.

The Tc-99 activity can also be calculated from mass-99 in the ICP-MS data. A comparison of the Tc-99 activity from the two measurements on the dried unwashed solids (Tc-99 method and ICP-MS) and the slurry value from ICP-MS is given in Table 3-10. Both measurements from the solids for all three PRFT batches are below the detection limit while the slurry activity is not, which suggests that Tc-99 in the PRFT is predominately in the supernate.

Table 3-7. Radionuclide Results for PRFT Batch 15 Dried Unwashed Solids

Radionuclides	µCi/g of Dried Solids*	pCi/g of Ti	% RSD	Analytical % Uncertainty	Method
Ni-59	< 2.25E-02	< 5.43E+04	NA	MDA	Ni-59/Ni-63
Ni-63	< 2.42E-02	< 5.83E+04	NA	Upper Limit	Ni-59/Ni-63
Se-79	<3.17E-03	<7.63E+03	NA	MDA	Se-79
Sr-90	1.15E+03	2.76E+09	4.2, 4	15%	Sr-90
Zr-93	7.31E-02	1.76E+05	2.6, 4	10%	ICP-MS
Nb-93m	6.42E-02	1.55E+05	NA	NA	Calculated
Tc-99	<6.44E-03	<1.55E+04	NA	MDA	Tc-99
Sn-126	< 1.11E-01	< 2.67E+05	NA	MDA	Cs-removed Gamma Scan
Cs-137	4.83E+02	1.16E+09	2.3, 4	5.0%	Gamma Scan
Sm-151	< 2.59E-02	< 6.23E+04	NA	Upper Limit	Pm-147/Sm-151
Th-229†	1.63E-05	3.92E+01	NA	24%	Th-229/230
U-233	2.25E-02	5.43E+04	3.3, 4	10%	ICP-MS
U-234	8.64E-02	2.08E+05	2.7, 4	10%	ICP-MS
U-235	2.41E-04	5.81E+02	2.9, 4	10%	ICP-MS
U-236	1.39E-03	3.35E+03	3.0, 4	10%	ICP-MS
U-238	1.46E-03	3.52E+03	1.8, 4	10%	ICP-MS
Np-237	7.75E-03	1.87E+04	3.1, 4	10%	ICP-MS
Pu-238	7.00E+01	1.69E+08	3.4, 4	4.7%	Pu-238/241
Pu-239	1.49E+00	3.60E+06	3.5, 4	10%	ICP-MS
Pu-240	7.79E-01	1.88E+06	3.0, 4	10%	ICP-MS
Pu-241	2.55E+01	6.15E+07	2.8, 4	15%	Pu-238/241
Pu-242	<2.37E-03	<5.71E+03	NA	MDA	ICP-MS
Am-241	1.02E-01	2.45E+05	6.6, 4	5.7%	AM/CM
Am-242m	< 2.25E-04	< 5.42E+02	NA	MDA	AM/CM
Am-243	< 1.12E-03	< 2.70E+03	NA	MDA	AM/CM
Cm-244	9.22E-03	2.22E+04	16, 4	12%	AM/CM
Cm-245	< 1.53E-03	< 3.69E+03	NA	MDA	AM/CM
Cm-246	< 5.86E-06	< 1.41E+01	NA	Upper Limit	AM/CM
Cm-247	< 2.50E-04	< 6.02E+02	NA	MDA	AM/CM
Cm-248	< 3.28E-07	< 7.90E-01	NA	Upper Limit	AM/CM
Cf-249	< 2.77E-04	< 6.66E+02	NA	MDA	AM/CM
Cf-251	< 1.60E-03	< 3.85E+03	NA	MDA	AM/CM

* Less than values represent the minimum detection limit value. † Th-229 only one of four replicates was above detection limit.

Table 3-8. Radionuclide Results for PRFT Batch 22 Dried Unwashed Solids

Radionuclides	µCi/g of dried Solids*	pCi/g of Ti	% RSD	Analytical % Uncertainty	Method
Ni-59	< 1.24E-02	< 3.01E+04	NA	MDA	Ni-59/Ni-63
Ni-63	< 1.85E-02	< 4.50E+04	NA	Upper Limit	Ni-59/Ni-63
Se-79	<3.42E-03	<8.31E+03	NA	MDA	Se-79
Sr-90	1.31E+03	3.17E+09	8.2, 4	15%	Sr-90
Zr-93	6.93E-02	1.68E+05	3.3, 4	10%	ICP-MS
Nb-93m	6.09E-02	1.48E+05	NA	NA	Calculated
Tc-99	<6.13E-03	<1.49E+04	NA	MDA	Tc-99
Sn-126	< 1.25E-01	< 3.03E+05	NA	MDA	Cs-removed Gamma Scan
Cs-137	1.13E+02	2.74E+08	5.8, 4	5%	Gamma Scan
Sm-151	< 2.69E-02	< 6.54E+04	NA	Upper Limit	Pm-147/Sm-151
Th-229	< 1.63E-05	< 3.95E+01	NA	MDA	Th-229/230
U-233	2.03E-02	4.94E+04	3.9, 4	10%	ICP-MS
U-234	9.21E-02	2.24E+05	4.1, 4	10%	ICP-MS
U-235	2.20E-04	5.34E+02	3.9, 4	10%	ICP-MS
U-236	1.26E-03	3.06E+03	3.8, 4	10%	ICP-MS
U-238	1.13E-03	2.75E+03	2.4, 4	10%	ICP-MS
Np-237	9.47E-03	2.30E+04	4.1, 4	10%	ICP-MS
Pu-238	8.27E+01	2.01E+08	2.7, 4	5%	Pu-238/241
Pu-239	1.46E+00	3.56E+06	3.4, 4	10%	ICP-MS
Pu-240	8.29E-01	2.02E+06	4.1, 4	10%	ICP-MS
Pu-241	3.07E+01	7.46E+07	4.4, 4	15%	Pu-238/241
Pu-242	< 2.50E-03	< 6.06E+03	NA	MDA	ICP-MS
Am-241	9.00E-02	2.19E+05	13, 4	6.9%	AM/CM
Am-242m	< 1.99E-04	< 4.83E+02	NA	MDA	AM/CM
Am-243	2.40E-03	5.82E+03	38, 4	22%	AM/CM
Cm-244	3.97E-03	9.65E+03	9.2, 4	16%	AM/CM
Cm-245	< 1.65E-03	< 4.01E+03	NA	MDA	AM/CM
Cm-246	< 4.00E-06	< 9.72E+00	NA	Upper Limit	AM/CM
Cm-247	< 2.64E-04	< 6.41E+02	NA	MDA	AM/CM
Cm-248	< 3.23E-07	< 7.84E-01	NA	Upper Limit	AM/CM
Cf-249	< 2.93E-04	< 7.12E+02	NA	MDA	AM/CM
Cf-251	< 1.58E-03	< 3.84E+03	NA	MDA	AM/CM

* Less than values represent the minimum detection limit value.

Table 3-9. Radionuclide Results for PRFT Batch 26 Dried Unwashed Solids

Radionuclides	µCi/g of dried Solids*	pCi/g of Ti	% RSD	Analytical % Uncertainty	Method
Ni-59	< 1.11E-02	< 2.66E+04	NA	MDA	Ni-59/Ni-63
Ni-63	< 2.08E-02	< 4.98E+04	NA	Upper Limit	Ni-59/Ni-63
Se-79	<3.30E-03	<7.91E+03	NA	MDA	Se-79
Sr-90	1.78E+03	4.27E+09	9.5, 4	15%	Sr-90
Zr-93	6.67E-02	1.60E+05	1.8, 4	10%	ICP-MS
Nb-93m	5.87E-02	1.41E+05	NA	NA	Calculated
Tc-99	<5.59E-03	<1.34E+04	NA	MDA	Tc-99
Sn-126	< 1.33E-01	< 3.19E+05	NA	MDA	Cs-removed Gamma Scan
Cs-137	9.61E+01	2.31E+08	2.2, 4	5%	Gamma Scan
Sm-151	< 3.44E-02	< 8.26E+04	NA	Upper Limit	Pm-147/Sm-151
Th-229	< 1.91E-05	< 4.58E+01	NA	MDA	Th-229/230
U-233	2.06E-02	4.94E+04	3.7, 4	10%	ICP-MS
U-234	1.03E-01	2.47E+05	1.9, 4	10%	ICP-MS
U-235	1.97E-04	4.73E+02	2, 4	10%	ICP-MS
U-236	1.26E-03	3.03E+03	2.4, 4	10%	ICP-MS
U-238	1.06E-03	2.54E+03	3.5, 4	10%	ICP-MS
Np-237	9.22E-03	2.21E+04	2.6, 4	10%	ICP-MS
Pu-238	1.02E+02	2.45E+08	4.4 4	5%	Pu-238/241
Pu-239	1.56E+00	3.76E+06	1.8, 4	10%	ICP-MS
Pu-240	9.26E-01	2.22E+06	2.4, 4	10%	ICP-MS
Pu-241	3.53E+01	8.47E+07	6.2, 4	15%	Pu-238/241
Pu-242	< 2.32E-03	< 5.57E+03	NA	MDA	ICP-MS
Am-241	6.67E-02	1.60E+05	7.9, 4	5.5%	AM/CM
Am-242m	< 1.37E-04	< 3.29E+02	NA	MDA	AM/CM
Am-243	< 4.91E-04	< 1.18E+03	NA	MDA	AM/CM
Cm-244	1.86E-03	4.45E+03	16, 4	20%	AM/CM
Cm-245	< 7.25E-04	< 1.74E+03	NA	MDA	AM/CM
Cm-246	< 1.78E-06	< 4.28E+00	NA	Upper Limit	AM/CM
Cm-247	< 1.19E-04	< 2.85E+02	NA	MDA	AM/CM
Cm-248	< 2.17E-07	< 5.20E-01	NA	Upper Limit	AM/CM
Cf-249	< 1.32E-04	< 3.18E+02	NA	MDA	AM/CM
Cf-251	< 7.66E-04	< 1.84E+03	NA	MDA	AM/CM

* Less than values represent the minimum detection limit value.

Table 3-10. Tc-99 Results Comparison

Analytical Method	Tc-99 in (pCi/g of Ti)		
	Tc-99 (Solids)	ICP-MS (Solids)	ICP-MS (Slurry)
PRFT Batch 15	<1.55E+04	<1.27E+04	7.52E+04
PRFT Batch 22	<1.49E+04	<1.35E+04	4.63E+04
PRFT Batch 26	<1.34E+04	<1.24E+04	3.87E+04

The U-235 enrichment is calculated using the following method documented in the DWPF Waste Acceptance Criteria:[15]

$$U-235\ Enrichment\ (Wt\ \%) = 100 \times \big(U-235+1.4(U-233)\big)/U \qquad (3)$$

where U is the summation of all uranium isotopes (U-233, U-234, U-235, U-236, U-238) and an analytical uncertainty of 2 sigma is accounted for in sample analysis. The U-235 enrichment of the MST/SS slurries is shown in Table 3-11, along with the enrichment values calculated by SRMC for StB 4, 5, and 7. There are slight differences between the PRFT enrichment values and the salt batch enrichment values which may be due to the unaccounted heels in the various tanks.

Table 3-11. U-235 Enrichment of the PRFT Slurry and StB Unfiltered Salt Solution

Sample	U-235 Enrichment
PRFT Batch 15	2.55%
StB 4	3.28%
PRFT Batch 22	2.93%
StB 5	2.43%
PRFT Batch 26	2.83%
StB 7	3.36%

The uranium and plutonium fissile isotopes are given in Table 3-12, in units of wt % total solids and microcuries per gram of total solids. These values are from ICP-MS of the acid digested slurries, except for Pu-241. The Pu-241 value is from the Pu-238/-241 method on the acid digestion of the PRFT dried unwashed solids. The Pu-241 value was converted to a slurry basis under the assumption that all the Ti within the PRFT slurry is in the insoluble MST/SS.

Table 3-12. Uranium and Plutonium Fissile Isotopes

		PRFT Batch 15	PRFT Batch 22	PRFT Batch 26
U-233	**Wt % Total Solids (% RSD)**	1.63E-04 (5.2)	1.66E-04 (1.6)	1.73E-04 (1.5)
	µCi/g of Total Solids	1.58E-02	1.61E-02	1.68E-02
U-235	**Wt % Total Solids (% RSD)**	7.52E-03 (2.7)	7.74E-03 (1.5)	7.14E-03 (0.7)
	µCi/g of Total Solids	1.63E-04	1.67E-04	1.54E-04
Pu-239	**Wt % Total Solids (% RSD)**	1.67E-03 (8.5)	1.85E-03 (1.5)	2.14E-03 (0.9)
	µCi/g of Total Solids	1.04E+00	1.15E+00	1.33E+00
Pu-241	**Wt % Total Solids (% RSD)**	1.76E-05 (2.8)	2.33E-05 (4.4)	2.74E-05 (6.2)
	µCi/g of Total Solids	1.81E+01	2.40E+01	2.83E+01

3.5 Slurry and MST/SS Solids Oxide Conversion

A wt % calcined solids can be calculated[16] for the three PRFT Batches. The elements detected at greater than 0.01 wt % in the slurry can be converted to oxides and summed, as shown in Table 3-13. In this calculation, it is assumed that these elements are converted to oxides and all mercury and anions such as nitrite, nitrate, and hydroxide are driven off in the calcining process. Therefore, this sum represents the mass of oxides (calcined solids) in 100 g of total dried solids. The sum was then used to calculate the wt % calcined solids (slurry basis). For PRFT Batch 15, the calculated wt % calcined solids is larger than the wt % insoluble solids. This appears to be due to the amount of sodium in the supernate; only ~45% of the sodium in the slurry is from MST for PRFT Batch 15. In comparison, for PRFT Batch 22 and 26, about 60% of the sodium is from MST.

The oxide results can be normalized and compared to current SSRT estimates[17] used in MAR assessments as shown in Appendix B. The ISS values from the PRFT batches are well below the SSRT 600 mg/L ISS estimate composition used, while the TiO_2 wt % of the SSRT composition estimates are lower than that of the PRFT batches. PRFT Batches 15, 22, and 26 are from salt batches that were processed with 0.4 g/L MST strikes at SWPF. SWPF has since switched to MST strikes as low as 0.1 g/L. Current results should be compared to PRFT batches post SWPF change to lower MST.

Table 3-13. Oxide Conversion for Elements in PRFT Batches

Element	Oxide	Oxide Conversion	Oxide (Wt % of Total Solids)		
			Batch 15	Batch 22	Batch 26
Al	Al_2O_3	1.8895	0.997	0.607	1.10
Ba	BaO	1.1165	0.0274	0.0274	0.0333
Ca	CaO	1.3992	0.660	0.547	0.710
Cd	CdO	1.1423	0.0205	0.0171	0.0155
Cr	Cr_2O_3	1.4616	0.0567	0.0518	0.0497
Fe	Fe_2O_3	1.4297	0.530	0.396	1.41
Mg	MgO	1.6583	0.0313	0.0236	0.0217
Mn	MnO	1.2912	0.0168	0.00574	0.112
Na	Na_2O	1.348	21.4	18.0	17.0
Ni	NiO	1.2726	0.0596	0.0510	0.132
S	SO_4	2.9959	0.476	0.478	0.412
Sr	SrO	1.1826	0.0134	0.00728	0.0109
Ti	TiO_2	1.6685	49.2	53.6	55.7
U	U_3O_8	1.1792	0.377	0.333	0.319
Zn	ZnO	1.2447	0.0373	0.0387	0.0599
Zr	ZrO_2	1.3507	0.0163	0.0185	0.0192
		Sum*	73.8	74.2	77.1
		Wt % Calcined Solids (Slurry Basis)	3.06	3.74	3.75

* The Sum represents the wt % calcined solids in the total solids. The wt % calcined solids (slurry basis), is calculated by multiplying the sum by the wt % total solids.

4.0 Conclusions

SRNL was tasked with the analyses of PRFT Batches 15, 22, and 26, which corresponds to material from the processing of StB4, StB5, and StB7, respectively. The elemental composition of the slurry and unwashed dried solids and the activity of the requested thirty-two radionuclides have been reported. The unwashed dried solids of the PRFT batches were found to be 86-87% MST. Additionally, the total sulfur values are well below the assumed 982 mg of sulfate/kg of PRFT slurry used in Material Tracking Program calculations.

5.0 References

1. *Waste Acceptance Product Specifications for Vitrified High-Level Waste Forms*; DOE/EM-0093, Rev. 2; Department of Energy - Office of Environmental Management: 1996.

2. Ray, J. W.; Culbertson, B. H.; Edwards, T. B.; Bibler, N. E.; Marra, S. L. *Reporting the Radionuclide Inventory of the DWPF Product*; WSRC-IM-91-116-4, Rev. 4; 2018.

3. Johnson, F. C. *Measurement Acceptance Region (MAR) Assessment Results Based on Confirmation Sample Sludge Batch 10 Projections*; SRNL-STI-2022-00647, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2023.

4. Silker, A. I.; Fogelman, P. E. *Impact Evaluation of Sludge Simulant in the Monosodium Titanate/Sludge Solids Stream on Processability at the Defense Waste Processing Facility*; SRMC-WSE-2022-00022, Rev. 0; Savannah River Mission Completion: Aiken, SC, 2022.

5. *Aqua Regia Dissolution of Sludge for Elemental Analysis*; Manual L16.1, Procedure ADS-2226, Rev. 10; Savannah River National Laboratory: Aiken, SC, 2013.

6. Trivelpiece, C. L.; Kubilius, W. P.; Diprete, D. P. *Determination of Reportable Radionuclides for Defense Waste Processing Facility (DWPF) Sludge Batch 9 (Macrobatch 11)*; SRNL-STI-2018-00680, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2019.

7. *Alkali Fusion Dissolutions of Sludge and Glass for Elemental and Anion Analysis*; Manual L16.1, Procedure ADS-2502, Rev. 7; Savannah River National Laboratory: Aiken, SC, 2013.

8. Pareizs, J. M.; Siegfried, M. J.; Martino, C. J.; Williams, M. S.; Stanfield, A. N.; Johnson, F. C.; Crawford, C. L. *Radioactive Waste Sludge Washing and Demonstration of the Nitric-Glycolic Acid Flowsheet for Sludge Batch 10 Qualification*; SRNL-STI-2022-00255, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2022.

9. Baxter, D. J. *Analysis of DWPF PRFT Samples*; X-TTR-S-00090, Rev. 2; Savannah River Mission Completion: Aiken, SC, 2021.

10. Hunter, S. C. *Task Technical and Quality Assurance Plan for the Analysis of DWPF Precipitate Reactor Feed Tank (PRFT) Samples*; SRNL-RP-2022-00295, Rev. 1; Savannah River National Laboratory: Aiken, SC, 2023.

11. *Salt Batch 6 and 7 Compliance with Material Tracking Program, Corrosive Species Evaluation, and Evaluating SE Additions at a Reduced Cs-137 Ci/gal*; X-CLC-S-00457, Rev. 7; 2023.

12. PRFT-1, Sample ID # 200029410, 200030224, and 200030688.

13. Hobbs, D. T.; Clearfield, A.; Visser, A. E.; Bridges, N. J. *Minor Actinide Separations Using Ion Exchangers or Ionic Liquids*; SRNL-STI-2011-00554, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2011.

14. Evington, M. C.; Click, D. R.; Hobbs, D. T., Sorption Behavior of Monosodium Titanate and Amorphous Peroxotitanate Materials Under Weakly Acidic Conditions. *Separation Science and Technology* **2009,** *45* (1), 66-72.

15. Ray, J. W. *Waste Acceptance Criteria For Raw Salt Solution, Sludge, and SWPF Salt Streams Transfers to DWPF*; X-SD-S-00001, Rev. 9; 2024.

16. Pareizs, J. M. *Analytical Results of the Tank 51H Sludge Batch 9 Qualification Sample (HTF-51-15-81)*; SRNL-L3100-2015-00142, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2015.

17. Stone, M. E. *SSRT Effluent Composition Estimate for SB9/10 MAR Assessments*; SRNL-L3300-2018-00063, Rev. 1; Savannah River National Laboratory: Aiken, SC, 2018.

Appendix A. Comparison of Activities between Slurry and Dried Unwashed Solids ICP-MS Data

Radionuclide	PRFT Batch 15 in pCi/g of Ti			PRFT Batch 22 in pCi/g of Ti			PRFT Batch 26 in pCi/g of Ti		
	Slurry (%RSD)	Solids (%RSD)	% Dif.	Slurry (%RSD)	Solids (%RSD)	% Dif	Slurry (%RSD)	Solids (%RSD)	% Dif
Zr-93	1.65E+05 (20)	1.76E+05 (2.6)	6.6	1.72E+05 (1.3)	1.68E+05 (3.3)	1.9	1.65E+05 (0.7)	1.60E+05 (1.8)	2.7
U-233	5.37E+04 (5.2)	5.43E+04 (3.3)	1.1	5.01E+04 (1.6)	4.94E+04 (3.9)	1.3	5.03E+04 (1.5)	4.94E+04 (3.7)	1.7
U-234	2.02E+05 (2.8)	2.08E+05 (2.7)	3.1	2.21E+05 (1.0)	2.24E+05 (4.1)	1.2	2.48E+05 (0.2)	2.47E+05 (1.9)	0.5
U-235	5.52E+02 (2.7)	5.81E+02 (2.9)	5.2	5.21E+02 (1.5)	5.34E+02 (3.9)	2.5	4.62E+02 (0.7)	4.73E+02 (2.0)	2.3
U-236	3.27E+03 (3.7)	3.35E+03 (3.0)	2.3	3.03E+03 (1.1)	3.06E+03 (3.8)	1.1	3.05E+03 (0.8)	3.03E+03 (2.4)	0.5
Np-237	1.82E+04 (4.5)	1.87E+04 (3.1)	2.8	2.25E+04 (1.5)	2.30E+04 (4.1)	2.3	2.18E+04 (0.7)	2.21E+04 (2.6)	1.4
U-238	3.54E+03 (4.9)	3.52E+03 (1.8)	0.6	2.82E+03 (1.8)	2.75E+03 (2.4)	2.7	2.57E+03 (0.4)	2.54E+03 (3.5)	0.8
Pu-239	3.52E+06 (8.5)	3.60E+06 (3.5)	2.2	3.57E+06 (1.5)	3.56E+06 (3.4)	0.4	3.99E+06 (0.9)	3.76E+06 (1.8)	5.9
Pu-240	1.80E+06 (4.9)	1.88E+06 (3.0)	3.9	1.98E+06 (0.8)	2.02E+06 (4.1)	1.8	2.24E+06 (1.3)	2.22E+06 (2.4)	0.8

Appendix B. Comparison of PRFT Batches to SSRT Composition Estimates

Oxide	Concentration (Wt %)				
	15*	22*	26*	SSRT Baseline	SSRT w/ 600 mg/L ISS
Al_2O_3	1.35	0.82	1.43	4.74	15.63
BaO	0.04	0.04	0.04	-	-
CaO	0.89	0.74	0.92	-	1.15
CdO	0.03	0.02	0.02	-	-
Ce_2O_3	-	-	-	-	0.17
Cs_2O	-	-	-	0.02	0.01
Cr_2O_3	0.08	0.07	0.06	-	-
Fe_2O_3	0.72	0.53	1.83	-	20.31
K_2O	-	-	-	0.51	0.24
MgO	0.04	0.03	0.03	-	0.29
MnO	0.02	0.01	0.15	-	7.37
Na_2O	28.92	24.23	22.07	46.31	22.44
NiO	0.08	0.07	0.17	-	0.78
P_2O_5	-	-	-	-	0.37
SiO_2	-	-	-	-	2.95
SO_4	0.64	0.64	0.53	0.25	1.34
SrO	0.02	0.01	0.01	-	-
ThO_2	-	-	-	-	0.71
TiO_2	66.59	72.27	72.22	48.17	23.34
U_3O_8	0.51	0.45	0.41	-	2.91
ZnO	0.05	0.05	0.08	-	-
ZrO_2	0.02	0.02	0.02	-	-

* Ce, K, P, Si, and Th were below detection in the PRFT slurries by ICP-AES and were not included in the oxide conversion.

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